



SIMATS Online Programming Lab

JAVA PROGRAMMING



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CO2-AT1-Assignment1

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QUESTIONS

Q1

Smart Document Editor using
String, StringBuilder and
StringBuffer

10 marks

**Q2**

Hospital Token Management
System using List Interface

10 marks

**Q3**

Airport Boarding Queue
Management using Queue
Interface

10 marks

**Q4**

Digital Certificate Verification
using Set Interface

10 marks

**Q5**

Smart Inventory Management
using Map Interface

10 marks



5/5 answered

Q1

Smart Document Editor using String, StringBuilder and StringBuffer

10
marks

Question 1 :

- 1.Read a paragraph from the user.
- 2.Count the number of characters, words, and sentences.
- 3.Replace every occurrence of a given word with another word.
- 4.Reverse each sentence individually.
- 5.Maintain editing history using StringBuffer.
- 6.Construct the final formatted document using StringBuilder.
- 7.Display the original document, editing history, and final document.

Requirement: Use String methods wherever appropriate.
Use StringBuilder for document construction and
StringBuffer for edit history.

Your Code

```
avd.util.Scanner;  
2  
lass SmartDocumentEditor {  
ic4static void main(String[] args) {  
Scanner scanner = new Scanner(System.in);
```

```

6
String paragraph = scanner.hasNextLine() ? scanner.nextLine() : scanner.nextLine();
String targetWord = scanner.hasNextLine() ? scanner.nextLine() : scanner.nextLine();
String replacementWord = scanner.hasNextLine() ? scanner.nextLine() : scanner.nextLine();
10
StringBuilder editHistory = new StringBuffer();
editHistory.append("History Log:\n");
editHistory.append("- Document created.\n");
14
int charCount = paragraph.length();
int wordCount = paragraph.trim().isEmpty() ? 0 : paragraph.trim().split(" ").length;
int sentenceCount = paragraph.trim().isEmpty() ? 0 : paragraph.trim().split("\\.\\?\\!").length;
18
String replacedParagraph = paragraph.replace(targetWord, replacementWord);
editHistory.append("- Replaced word '").append(targetWord).append("' with '").append(replacementWord).append("'.\n");
25
StringBuilder finalDocument = new StringBuilder();
String[] sentences = replacedParagraph.split("\\.\\?\\!");
28
for (String sentence : sentences) {
    if (!sentence.trim().isEmpty()) {
        StringBuilder tempSentence = new StringBuilder(sentence.trim().split(" ").length);
        finalDocument.append(tempSentence.reverse().toString().trim() + ".\n");
    }
}
34
editHistory.append("- Reversed each sentence and added period.\n");
36
System.out.println("\n --- ORIGINAL DOCUMENT ---");
System.out.println(paragraph);
39
System.out.println("\n --- DOCUMENT STATISTICS ---");
System.out.println("Character Count : " + charCount);
System.out.println("Word Count : " + wordCount);
System.out.println("Sentence Count : " + sentenceCount);
44
System.out.println("\n -- EDITING HISTORY --");
System.out.println(editHistory.toString());
47
System.out.println("\n --- FINAL DOCUMENT ---");
System.out.println(finalDocument.toString().trim());
50

```

```
51 mer.close();  
52  
53  
54
```

Input

stdin input

Output

--- ORIGINAL DOCUMENT ---

--- DOCUMENT STATISTICS ---

Character Count : 0

Word Count : 0



Run



Save